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Aggregate Data, Ecological Regression, and Voting Transitions

PHILIP J. BROWN and CLIVE D. PAYNE*

Voting data typically comprise the marginal distributions of votes cast at two successive elections. The fact that these obtain separately for many voting precincts or areas enables one to estimate the actual transition probabilities for movements between the options available to the voter at each election. An aggregated compound multinomial model is proposed. This allows log-linear dependence on various covariates and specifies a simple and illuminating structure of random effects. The information in such aggregated data is compared with that which would obtain had all of the transitions been observed. The model is applicable to many other types of aggregated data and meets many of the difficulties inherent in "ecological regression." It is illustrated with an analysis of the British European election of 1984.

KEY WORDS: Compound multinomial; Information; Markov chains; Compartment models; Matrix derivatives; Elections.

1. INTRODUCTION

Suppose that T_{ijk} denotes the number of individuals in the area k (e.g., British constituency) ($k = 1, \dots, K$) who chose option i ($i = 1, \dots, I$) in the first election and option j ($j = 1, \dots, J$) in the second election. A full set of options would include the parties standing, abstention, categories of people at the second election who have come onto the electoral register between elections, and those at the first election who leave the register between elections. Unfortunately for statistical analysis, T_{ijk} are not observed. Indeed such linkage of individual behavior across elections would be counter to the essential confidentiality of the ballot. However, the marginal totals $X_{ik} = \sum T_{ijk}$ and $Y_{jk} = \sum T_{ijk}$ are observed. Data on the socioeconomic and political characteristics of the areas are also usually available. Statistical analysis of such aggregated data has enjoyed a lively debate in the sociological literature. The near mortal blow struck by Robinson (1950), documenting the dangers of "ecological correlation" in which aggregate level correlations tend to be much larger than individual level correlations, was treated by Goodman (1953, 1959) by using "ecological regression" and subsequently by Stokes (1969), Shively (1969), Przeworski (1974), Hanushek, Jackson, and Kain (1974), Crewe and Payne (1976), and Wright (1977), who emphasized the need for careful modeling in terms of covariates, especially those covariates that define the context of the individual in the particular aggregate unit.

Important contributions to solving the problem of estimation of voter transition rates were also given in the statistical literature by Hawkes (1969), Irwin and Meeter (1969), Miller (1972), and McCarthy and Ryan (1977). Irwin and Meeter as well as

McCarthy and Ryan analyzed Goodman's ecological regression model

$$E(Y_{jk}) = \sum p_{ij} X_{ik}, \quad j = 1, \dots, J - 1, \quad (1.1)$$

using quadratic programming to limit the range of p_{ij} , the estimated probability that an individual choosing option i at election 1 will subsequently choose option j , to $p_{ij} \geq 0$, $\sum p_{ij} \leq 1$. Here Y_{jk} is completely determined by the other $(J - 1) Y_{jk}$ and the electorate, X_{ik} . Irwin and Meeter suggested a weighted least squares analysis, whereas Hawkes, after discarding an ill-fitting multinomial model, also worked with (1.1) but with the $(J - 1) \times (J - 1)$ covariance matrix $(\sqrt{X_{ik}})\Sigma$ with Σ unknown. Miller, like Stokes earlier, recognized that inadmissible p_{ij} might be indications of a poorly specified model, and he attempted to remedy this by replacing p_{ij} by $a_{ij} + b_{ij}z_{ik}$, where z_{ik} was the Conservative percentage of the Conservative plus Labour two-party vote in the k th constituency in the United Kingdom, calling this variable the *partisanship* of the constituency. Persistent inadmissible estimates were pulled into the $(0, 1)$ interval by a form of ridge regression.

The existing methodology and that which we will develop in this article is applicable to a wide variety of data structures. One type of application is that in which individuals or objects have been exclusively and exhaustively classified at two time instants and this has been repeated many times either spatially, as in election studies, or temporally, as in economic models [e.g., brand switching over time in Telser (1962)] or as in biological models [e.g., compartment residency of Kodel and Matis (1976) and Kalbfleisch, Lawless, and Vollmer (1983)]. Within the areal election studies, further classification is possible. The two separate elections may be from primary and runoff elections with a subset of candidates in common, as in Irwin and Meeter (1969). The models are also applicable to the analysis of data on votes for two offices at the same election. Interest might concentrate on $\sum_{i \neq j} p_{ij}$, the extent of split-ticket voting, as in Cowart (1974).

A second type of application arises when objects are separately classified on a number of variates and the marginal distributions are available for a set of areal units at one time point. The aim here is to estimate the internal cells in the cross-classifications. Examples are race and literacy (Robinson 1950) and class and vote (Crewe and Payne 1976). Most analysis of aggregate data has involved data of this type.

Many social and political researchers have been reluctant to use aggregate data, given the methodological difficulties. The primary instrument has been the social survey. Survey data are much more expensive to collect than the collation of aggregate data from official sources. Surveys also have their own special difficulties of nonresponse, selection, and response bias. In the

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voting context it is well known that voting surveys typically underestimate vote switching, abstention, and third-party votes (see Himmelweit, Jaeger, and Stockdale 1978). Good methods for the analysis of aggregate data are needed to redress the imbalance between the efforts devoted to these two approaches.

In Section 2 the proposed model is defined and contrasted with models in the literature. Section 3 gives a large sample likelihood analysis of data from this model. Some insights into the loss of information inherent in aggregating are given in Section 4. Finally, the methodology is illustrated on some British data in Section 5. More data analysis and examples are available in Firth (1982).

2. THE AGGREGATED COMPOUND MULTINOMIAL MODEL

We begin by imagining a single row of the $I \times J$ category contingency table. This i th row has J entries corresponding to the J options in the current election open to the x_i individuals who previously took the i th option. These entries are unobserved transitions, and suppressing the row label for the time being, they may be denoted by $T_1, \dots, T_J$, where since these add up to x , the observed row total, attention may be restricted to the first $(J - 1)$ of these.

Following Mosimann (1962) or Johnson and Kotz (1969, chap. 11) let us suppose that the random transitions have a compound multinomial in J dimensions. That is, the random variables $(T_1, \dots, T_{J-1}) = \mathbf{T}^T$ with index x , a known integer, and J parameters $\alpha > 0$, $\mathbf{p} = (p_1, \dots, p_{J-1})^T$, $\sum p_j \leq 1$, $p_j \geq 0$ have a compound multinomial, denoted by $C(\mathbf{p}, \alpha; x)$. Here, implicitly $T_J = x - \sum T_j$, $p_J = 1 - \sum p_j$, and the parameterization is such that p_j is the expected proportion of the x individuals taking option j . This distribution has moments

$$E(\mathbf{T}) = x\mathbf{p}, \quad \text{cov}(\mathbf{T}) = [(x + \alpha)/(1 + \alpha)]x\mathbf{\Omega}, \quad (2.1)$$

where

$$\mathbf{\Omega} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^T. \quad (2.2)$$

For insight into the role of the parameter α , note that the distribution may be characterized as a Dirichlet mixture of a standard multinomial, allowing an inflation in variance over the usual multinomial, and the single parameter α determines this additional variance. The distribution may obtain from two stages. In the first stage, $T_1, \dots, T_J$ is multinomial with index x and probabilities $\phi_1, \dots, \phi_J$, $\sum \phi_j = 1$. The ϕ parameters are in turn obtained as a sample from a Dirichlet distribution, the J category generalization of a beta, with probability density function proportional to

$$\prod \phi_j^{\alpha_j - 1} (1 - \sum \phi_j)^{\alpha - 1},$$

where $\alpha_j = \alpha p_j$, $j = 1, \dots, J - 1$, and $\alpha_J = \alpha - \sum \alpha_j$. It will be useful to recall from the properties of the Dirichlet given in Wilks (1962) that

$$E\phi = \mathbf{p}, \quad \text{var}(\phi_j) = p_j(1 - p_j)/(1 + \alpha), \\ \text{cov}(\phi_j, \phi_m) = -p_j p_m / (1 + \alpha).$$

Note that we have chosen the parameterization $(\mathbf{p}, \alpha)$ rather than the customary $(\alpha_1, \dots, \alpha_J)$, where $\alpha = \sum \alpha_j$ and $p_j = \alpha_j / \alpha$. The model is a random effects model with the rather

economical use of a single parameter α for variability. In fact the larger the value of α , the less the variability of transition probabilities between areas; hence it is better to think of α as the precision parameter. When $\alpha \rightarrow \infty$ the variability tends to zero and the distribution of $\mathbf{T}$ reverts to the multinomial.

Typically $\mathbf{p}$ will not be constant across areal units or districts, indexed by k , but may be specified as a function of an unknown parameter $\boldsymbol{\beta}$, conditional on known covariates, where a natural parameterization is log-linear; that is,

$$\log(p_{jk}/p_{Jk}) = \beta_{j0} + \sum \beta_{jl} z_{ilk}, \\ j = 1, \dots, J - 1, k = 1, \dots, K, \quad (2.3)$$

with $z_1, \dots, z_L$ known covariates. Thus the systematic effects are modeled on covariates through the parameters $\boldsymbol{\beta}$, the random effects through the parameter α .

With this definition of a compound multinomial, an aggregated compound multinomial is defined as follows: Suppose $\mathbf{T}_i = (T_{i1}, \dots, T_{iJ-1})$ are independent $C(\mathbf{p}_i, \alpha_i; x_i)$, $i = 1, \dots, I$; that is, each row of the $I \times J$ table is compound multinomial with parameters differing by row. Then the margin formed from adding rows to give $\mathbf{Y} = (Y_1, \dots, Y_{J-1})^T$, where $\mathbf{Y} = \sum \mathbf{T}_i$, is aggregated compound multinomial and denoted by $A(\mathbf{P}, \boldsymbol{\alpha}; \mathbf{x})$, where the $I \times (J - 1)$ matrix $\mathbf{P} = (p_1^T, \dots, p_I^T)$ is the matrix of expected transition probabilities for an individual and $\boldsymbol{\alpha}^T = (\alpha_1, \dots, \alpha_I)$, $\alpha_i > 0$, $i = 1, \dots, I$, are the I different row precisions. The known column margin is $\mathbf{x} = (x_1, \dots, x_I)$. If $\mathbf{Y}$ is $A(\mathbf{P}, \boldsymbol{\alpha}; \mathbf{x})$, then its moments are

$$E(\mathbf{Y}) = \mathbf{P}^T \mathbf{x} \quad (2.4)$$

and

$$\text{cov}(\mathbf{Y}) = \text{diag}(\mathbf{P}^T \mathbf{w}) - \mathbf{P}^T \text{diag}(\mathbf{w}) \mathbf{P}, \quad (2.5)$$

with $\mathbf{w} = (w_1, \dots, w_I)^T$ and $w_i = x_i (x_i + \alpha_i) / (1 + \alpha_i)$, $i = 1, \dots, I$. The sample $\mathbf{Y}_1, \dots, \mathbf{Y}_K$ of K areas are assumed independent $A(\mathbf{P}_k, \boldsymbol{\alpha}; \mathbf{x}_k)$, $k = 1, \dots, K$, so that

$$E(\mathbf{Y}_k) = \mathbf{P}_k^T \mathbf{x}_k \quad (2.6)$$

and

$$\text{cov}(\mathbf{Y}_k) = \text{diag}(\mathbf{P}_k^T \mathbf{w}_k) - \mathbf{P}_k^T \text{diag}(\mathbf{w}_k) \mathbf{P}_k = \mathbf{\Gamma}_k, \quad (2.7)$$

defining $\mathbf{\Gamma}_k$. Here α is envisaged constant over k . Typically $\mathbf{P}_k$ will not give separate parameter matrices but will be related through log-linear dependence on known covariates z_{ilk} as in (2.3) so that

$$\log(p_{ijk}/p_{Jk}) = \beta_{j0} + \sum \beta_{jl} z_{ilk}. \quad (2.8)$$

Although the aggregated multinomial is not new, the aggregated compound multinomial is new and presents distinct advantages over the aggregated multinomial. In the special case of $\alpha \rightarrow \infty$, then $w_i \rightarrow x_i$ and the aggregated compound multinomial specializes to the aggregated multinomial. MacRae (1977) developed corresponding special cases of formulas (2.6) and (2.7), noting that (2.7) corrected an error in Lee, Judge, and Zellner (1970). She also suggested the log-linear dependence of $\mathbf{P}_k$ on covariates. Note that (2.6) corresponds to the regression model (1.1) so that the difference between our approach and that of, say, Irwin and Meeter (1969) or McCarthy and Ryan

(1977) is (a) the use of the special error structure (2.7) and (b) log-linear dependence of $\mathbf{P}_k$ as in (2.3), on covariates allowing varying transition probabilities within the range (0, 1) without need for restrictions on β_{j0}, β_{jl} in (2.3). Feature (a) is particularly important, as it allows an internal measure of goodness of fit, allows us to examine residuals in absolute terms as well as when plotted against possible covariates, and quantifies the extent to which area transitions vary randomly about the I rows of $\mathbf{P}_k$ through the precision parameters α . Hawkes (1969), realizing that election data involve more variability than a multinomial would suggest, declined to develop a random effects model on the grounds that it would involve too many extra parameters. Our random effects model need add merely I extra parameters.

3. MODEL ESTIMATION

The data, as we have seen, consist of K pairs of margins of $K I$ -option $\times J$ -option contingency tables. In addition there are L explanatory variables that vary between the K areal units.

To estimate the unknown parameters from these data, first it is natural to attempt a likelihood analysis of the aggregated compound multinomial model described before. If the actual transitions T_{ijk} had been observed, then the likelihood could be computed as the product over i, k of the compound multinomial likelihoods for $T_{ijk} = t_{ijk}$ observed as a function of the unknown $\mathbf{P}_k, \alpha$, and known x_{ik} . Since only

$$\mathbf{Y}_k = \sum \mathbf{T}_{ik} \quad (3.1)$$

is observed, however, the product of compound multinomial likelihoods must be summed over all $\mathbf{T}_{ik}$ satisfying (3.1). With $\mathbf{Y}_k$ of the order of several thousand, this is an awesome computational task even with modern computing power. At least two alternatives are possible.

1. Given that $\mathbf{Y}_k$ is large, one may assume that $\mathbf{Y}_k$ are independent $(J - 1)$ -variate normal with mean vector $\mathbf{P}_k^T \mathbf{x}_k$ and covariance matrix Γ_k , where Γ_k is given by (2.7).

2. Regard the problem as one of missing information and apply the EM algorithm of Dempster, Laird, and Rubin (1977).

The latter, although appealing in its relative simplicity, suffered from such slow convergence in the multinomial case that it was impossible to tell whether it had converged (see Firth 1982). The former large sample approach is adopted here.

The large sample log-likelihood, l , may be written as $\sum l_k$, where apart from an additive constant,

$$-2l_k = \log |\Gamma_k| + (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k), \quad (3.2)$$

where $\mathbf{H}_k = (\Gamma_k)^{-1}$. A good numerical optimization procedure for such a nonlinear log-likelihood would seem to be a quasi-Newton method (e.g., see Chambers 1973), involving the specification of theoretical first derivatives.

Appendix A calculates analytical first derivatives. These are, first, with respect to $\mathbf{P}_k$,

$$\begin{aligned} -2\partial l / \partial \mathbf{P}_k &= \mathbf{w}_k^T \mathbf{D}(\mathbf{H}_k) - 2 \text{diag}(\mathbf{w}_k) \mathbf{P}_k \mathbf{H}_k \\ &\quad - \mathbf{w}_k \{ \mathbf{H}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k) \}^{2T} \\ &\quad + 2 \text{diag}(\mathbf{w}_k) \mathbf{P}_k \mathbf{H}_k \\ &\quad \times (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k) (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k \\ &\quad - 2 \mathbf{x}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k, \end{aligned} \quad (3.3)$$

where $\{\mathbf{a}\}^{2T} = (a_1^2, \dots, a_{j-1}^2)$ and $\mathbf{D}(\mathbf{A})$ is a diagonal matrix with j th diagonal element a_{jj} when $\mathbf{A} = (a_{ij})$. In addition, for derivatives with respect to the α parameters, it is more convenient to work with the new reparameterization, $\theta_i = 1 / (1 + \alpha_i)$, so that following the definition of w after (2.5),

$$w_{ik} = x_{ik} [1 + (x_{ik} - 1)\theta_i].$$

Then

$$\begin{aligned} -2\partial l_k / \partial \theta_i &= x_{ik}(x_{ik} - 1) [\text{trace}(\mathbf{H}_k \mathbf{\Omega}_{ik}) \\ &\quad - (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k \mathbf{\Omega}_{ik} \mathbf{H}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)], \end{aligned} \quad (3.4)$$

where $\mathbf{\Omega}_{ik} = \text{diag}(\mathbf{p}_{ik}) - \mathbf{p}_{ik} \mathbf{p}_{ik}^T$ and $\mathbf{p}_{ik}$ is the i th column of $\mathbf{P}_k^T$, $i = 1, \dots, I$.

As a check on (3.3) and (3.4) one may verify that their expectations are zero; that is, $E(\partial l / \partial \mathbf{P}_k) = 0$ ($I \times (J - 1)$) and $E(\partial l / \partial \theta) = 0$ ($I \times 1$). This is a standard property of the log-likelihood.

Since the matrices of transition parameters, $\mathbf{P}_k$, $k = 1, \dots, K$, are actually functions of but a few unknown parameters, the β_{ijl} of (2.8), to obtain the derivative of the log-likelihood with respect to these unknown β parameters, note that

$$\partial l_k / \partial \beta_i^T = (\partial l_k / \partial \mathbf{p}_{ik}) \partial \mathbf{p}_{ik} / \partial \beta_i^T = \sum (\partial l_k / \partial p_{ijk}) \partial p_{ijk} / \partial \beta_i^T, \quad (3.5)$$

where we make the simplifying assumption that β_i , the β components of the i th row $\mathbf{p}_{ik}^T$ of $\mathbf{P}_k$, are not involved in any other rows of $\mathbf{P}_k$. For example, we do not envisage that any covariate will have the same effect in different rows of the $I \times J$ two-way tables. Furthermore, with each row of $\mathbf{P}_k$ parameters as (2.8), the requisite derivatives are

$$\begin{aligned} \partial p_{jk} / \partial \beta_j^0 &= p_{jk}(1 - p_{jk}); \\ \partial p_{mk} / \partial \beta_j^0 &= -p_{mk} p_{jk}, \quad m \neq j, \\ \partial p_{jk} / \partial \beta_{jl} &= z_{lk} p_{jk}(1 - p_{jk}), \\ \partial p_{mk} / \partial \beta_{jl} &= -z_{lk} p_{mk} p_{jk}, \quad m \neq j, \end{aligned} \quad (3.6)$$

for $j = 1, \dots, J - 1$, $l = 1, \dots, L$, and where for clarity the additional i suffix has been omitted from each variable. In vector form this becomes

$$\partial \mathbf{p}_{ik} / \partial \beta_i = (\mathbf{\Omega}_{ik}, z_{i1k} \mathbf{\Omega}_{ik}, \dots, z_{iLk} \mathbf{\Omega}_{ik}). \quad (3.7)$$

In the special case in which $\alpha \rightarrow \infty$, the aggregated multinomial model, iterative reweighted least squares have been used to maximize the likelihood [see Hawkes (1969), MaCrae (1977), Kalbfleisch et al. (1983), and Thomson and Baker (1981)]. In this multinomial case,

$$\Gamma_k = x_{\cdot k} [\text{diag}(\mathbf{P}_k^T \mathbf{u}_k) - \mathbf{P}_k^T \text{diag}(\mathbf{u}_k) \mathbf{P}_k],$$

where $u_{ik} = x_{ik} / x_{\cdot k}$ with $x_{\cdot k} = \sum x_{ik}$. Hence assuming the multinomial model true $\mathbf{Y}_k - \mathbf{P}_k^T \mathbf{x}_k$ is $O_p(\sqrt{x_{\cdot k}})$ and the last term of the derivative of the likelihood contribution in (3.3),

$$-2 \mathbf{x}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k$$

is dominant and of order $\sqrt{x_{\cdot k}}$ close to the maximum of l , whereas all other terms are only of order 1 close to the maximum. The first and third terms of (3.3) are of order 1 even when the parameter vector is far from the maximizing value.

In consequence the contribution to (3.3) from the determinant in (3.2) is insignificant, justifying iteratively reweighted least squares. One bizarre property of that procedure, fortuitous perhaps in compartment models (see Kalbfleisch et al. 1983), is that the solution is invariant to changes of scale $\mathbf{y}_k \rightarrow c_k \mathbf{y}_k$, $\mathbf{x}_k \rightarrow c_k \mathbf{x}_k$, where c_k is a scaling constant for the k th table. This is not true of our approximate maximum likelihood estimator maximizing (3.2).

In the aggregated compound multinomial model it is essential to retain the log determinant contribution in (3.2). Iteratively reweighted least squares fail to provide any estimates of α . In addition, if α is not large, Γ_k is of order x_k^2 so that all of the terms in (3.3) are of the same order 1. Thus iteratively reweighted least squares and maximum likelihood do not correspond asymptotically when α is specified by some other method.

Hypotheses about the underlying β parameters may be tested in the usual way by differencing the log-likelihoods from unrestricted β and β constrained by the hypothesis, then applying the asymptotic theory that -2 times this is chi-squared on ν degrees of freedom, where ν is the difference in the number of free parameters in the two models. Additionally, for residual analysis, one may use standardized components, $(\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k) / \mathbf{H}_k$. They may be plotted against possible covariates, an important model-building technique, and also examined for outliers. Moreover, as a measure of goodness of fit,

$$\sum (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k)^T \mathbf{H}_k (\mathbf{y}_k - \mathbf{P}_k^T \mathbf{x}_k) \quad (3.8)$$

will be asymptotically chi-squared on $[K(J - 1) - \#(\text{free parameters})]$ degrees of freedom when the aggregated multinomial model is true.

An added bonus of the quasi-Newton method of likelihood maximization is that it gives an estimated-second-derivatives matrix as output. The inverse of this provides an estimated covariance matrix of the α , β and hence allows large sample confidence intervals to be specified. These may also be translated back to the α and $\mathbf{P}$ matrix, since $\xi = (\mathbf{p}_1^T, \dots, \mathbf{p}_I^T, \alpha^T)$; $\text{cov}(\xi) = \mathbf{A}\mathbf{V}\mathbf{A}^T$, where

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix},$$

and $\mathbf{A}_{11} = (a_{mn})$ with $a_{mn} = \partial p_m / \partial \beta_n$ and p_m the m th element of ξ , $\mathbf{A}_{22} = \mathbf{I}$, the $(I \times I)$ identity matrix, and $\text{cov}(\gamma) = \mathbf{V}$, where $\gamma = (\beta_1^T, \dots, \beta_I^T, \alpha^T)^T$.

Analytical expressions for the expected second derivatives of the log-likelihood parameterized in terms of $\mathbf{P}_k$ and θ are given in Appendix B. They are used in the next section to compare the information content of aggregated data as compared to that which would have obtained had the transitions been observed.

Details of a FORTRAN program for fitting the aggregated compound multinomial using the Numerical Algorithms Group (Oxford, U.K.) minimization routine E04KBF are available from the authors.

4. COMPARISON OF FULL AND AGGREGATED INFORMATION

This section theoretically compares the information matrix for the aggregated compound multinomial with that which would have obtained had all within-table transitions been observed.

Since large sample confidence intervals can be derived from the inverse of the information matrix, we are implicitly comparing width of confidence regions.

The information matrix for the compound multinomial is calculated in Appendix B. To compare this with the information that would have obtained had all transitions been observed, we specialize for simplicity to $\mathbf{P}_k = \mathbf{P}$, the case of constant transitions, and α , the Dirichlet variability parameter, fixed, when the information matrix from Appendix B, provided α is reasonably large, is dominated by the third term of (B.1)—that is,

$$E(-\partial^2 \ell / \partial \mathbf{p}_i \partial \mathbf{p}_m^T) = \sum x_{ik} x_{mk} \mathbf{H}_k, \quad (4.1)$$

where $\mathbf{p}_i^T$, $\mathbf{p}_m^T$ are the i th and m th rows of $\mathbf{P}$. Denoting the full likelihood as l_0 ,

$$E(-\partial^2 l_0 / \partial \mathbf{p}_i \partial \mathbf{p}_m^T) = \sum (x_{ik}^2 / w_{ik}) \Omega_i^{-1} \quad (4.2)$$

with $\Omega_i = \text{diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^T$; $E(\partial^2 l_0 / \partial \mathbf{p}_i \partial \mathbf{p}_m^T) = 0$, $i \neq m$; and (4.2) obtains from setting $I = 1$ in the aggregated model. When $\alpha \rightarrow \infty$ the compound multinomial degenerates to the multinomial, and when $J = 2$ comparison of (4.1) and (4.2) becomes relatively simple, since the p vectors in these equations become scalars, as Ω_i and Γ_k and hence $\mathbf{H}_k$. Thus

$$E(-\partial^2 l / \partial \mathbf{p} \partial \mathbf{p}^T) = \sum \mathbf{x}_k \mathbf{x}_k^T / \Gamma_k \quad (4.3)$$

and

$$E(-\partial^2 l_0 / \partial \mathbf{p} \partial \mathbf{p}^T) = \text{diag}(\mathbf{a}), \quad (4.4)$$

where $a_i = x_i / p_i(1 - p_i)$.

The variances of estimators of linear functions of $\mathbf{p}$ may be contrasted using the inverses of these information matrices. Typically certain linear combinations are well estimated by the aggregated data whereas others are not. For a detailed general comparison in Markov chains, see Lawless and McLeish (1984) and Rosenqvist (1985).

5. VOTING TRANSITIONS: AN ILLUSTRATION

5.1 The Problem

We estimate voter transition probabilities for the British election to the European Parliament in June 1984 from the previous British General Election in June 1983. The European election was fought in 79 constituencies formed from nonoverlapping aggregates of from 8 to 10 of the 650 (Westminster Parliament) constituencies for the preceding General Election. Thus we were able to construct a notional set of results for a 1983 election in the European constituencies, and this was used as a base for the estimation of transitions. We restrict attention to all of the 66 English constituencies. These were fought by the three major parties [Conservative, Labour, and the Liberal-Social Democratic Party (SDP) Alliance] at both elections. Small minor parties also stood in most of these constituencies.

Most commentators considered the European election essentially an opportunity for electors to vote on national party lines rather than on European issues. Thus it was viewed as a referendum on the performance of the parties in the British parliament—in effect a mini British General Election but one with low salience to the electorate.

The final results for both elections are given in Table 1. The main feature to note is that there was a much higher degree of abstention in 1984 and there is interest in the degree of differ-

Table 1. Shares of the Electorate in England

Party	1983 General Election	1984 European Election
Conservative	33.3	13.8
Labour	19.5	11.2
Alliance	19.2	6.3
Others	.5	.3
Abstentions	27.5	68.4

ential abstention. The Labour party increased its share of the votes cast from 26.9% to 35.5%, recovering from a poor performance in 1983. The Conservative share of the vote fell slightly from 46.0% to 43.7%, whereas the Alliance dropped 6.3 percentage points to 20.1%. Thus interest also focuses on the sources of recovery for Labour and decline for the Alliance.

A special feature of the 1984 election was that the Conservative government was about to introduce legislation to abolish the predominantly Labour-controlled Metropolitan Councils, a tier of local government. This was an issue emphasized by the Labour party, and we use the model to test the hypothesis that residence in a Metropolitan county affected voting behavior.

5.2 The Data

The data available are not ideal for our purposes, and some adjustments have to be made to allow analysis to proceed. The adjustments imply the following assumptions about the voting behavior of different types of elector.

1. The data available do not include the numbers of the voters in the 1983 election in a constituency who did not vote in 1984 for reasons of death or emigration. Similarly they do not include the numbers of voters who either came of age between the two elections or moved into the constituency between the two elections. We follow the scheme proposed by Hawkes (1969), who scaled the marginal totals for the first election to have the same relative proportions as the marginals for the second. This modification essentially implies the assumption that the voters leaving the electoral register through death or emigration or joining it through coming-of-age or immigration have the same voting profile as the rest of the voters in the constituency.

2. The 1983 election was fought on the component Westminster constituencies, and individual voters may have been influenced by the political characteristics of their particular constituency, such as its marginality, or by the personality of a particular candidate. Thus aggregation to the European constituencies assumes that these local effects are averaged out. Detailed examination of residuals may identify some European constituencies in which divergent voting patterns in a component Westminster constituency in 1983 have had a significant effect on voting transitions in the European constituency.

3. The primary source of explanatory variates is the 1981 census data for the European constituencies. These data are three years out of date, and we have to assume that the socio-economic characteristics of the constituencies have remained unchanged. However, significant changes in the composition of the individual electorates may have taken place in the period 1981–1984.

4. Most constituencies have minor parties standing. In the English constituencies these parties gained very small shares of the electorate, and we have not thought it worthwhile to

consider them as a separate voting option. Accordingly, we combine their votes with abstentions, and the data structure on which we base our analysis is the set of marginal totals for Conservative, Labour, Alliance, and the rest (abstentions plus votes for minor parties) for each of the elections. Pooling minor party votes with abstentions implies that the 1984 voting profile of supporters of minor parties in 1983 is the same as that of 1983 abstentions. This assumption is unlikely to be true, but its impact is likely to be small.

A full analysis of the British European election would also include the four Welsh constituencies and eight Scottish constituencies in which significant nationalist parties also stood. Our model can be extended to deal with an incomplete pattern of parties standing in the full set of constituencies.

Data problems 1 and 2 are common to the analysis of successive elections in most countries, whereas problem 3 is specific to the particular type of election being considered here. There are many other applications of the model in which the available data are more suitable. This particularly applies to data with marginals based on the same population (for example, split-ticket voting) or to data intended to estimate voting profiles for different socioeconomic groups of voters at a single election.

5.3 Model Fitting

We treat the model fitting as a sequential process. We start with the constant probability model and add explanatory variates guided by our substantive hypotheses about voting behavior in British elections, by an examination of the goodness of fit of each model, and by a detailed analysis of the correlation residuals with candidate covariates. A model sequence is given in Table 2.

Two measures of fit are proposed to complement the analysis of the decrease in minus twice the log-likelihood of assessing fit. Measure 1 is defined as the mean of the absolute differences between fitted and observed percentage share for each option:

$$\sum_k |\hat{P}_{jk} - P_{jk}| / K \quad \text{for option } j.$$

Measure 2, the percentage fit, is defined as

$$1 - \frac{\left[\sum_{k=1}^K \sum_{j=1}^J (Y_{jk}/X_{\cdot k} - \hat{P}_{jk})^2 \right]}{\left[\sum_{k=1}^K \sum_{j=1}^J (Y_{jk}/X_{\cdot k} - \hat{P}_{ojk})^2 \right]}, \quad (5.1)$$

where $\hat{P}_{jk}$ is the fitted proportion for option j of the electorate $X_{\cdot k}$ in the k th constituency and $\hat{P}_{ojk}$ is a baseline fitted model assuming a constant proportional change, dependent on option j , where with the same set of options between the two elections, $\hat{P}_{ojk} = [X_{jk}/X_{\cdot k}]D_j$ and $D_j = Y_j/X_j$. Thus the ratio in (5.1) is

$$\frac{\text{residual sum of squares from fitted model}}{\text{total sum of squares}},$$

where *total* denotes a baseline proportional change model.

The model for P_k is

$$\log(P_{ijk}/P_{ilk}) = \beta_{ijo},$$

$$j = 1, \dots, 3, J = 4, i = 1, \dots, 4,$$

Table 2. The Fitted Models

Model	df	Measure 1				Measure 2, % Fit	Change in -2 log-likelihood	df	α Parameters			
		Con.	Lab.	All.	Rest				α_1	α_2	α_3	α_4
Constant	182	.9	.9	.8	2.1	83			850	83	81	1
+ Turnover	170	.7	.8	.7	1.7	89	63.6	12	1,576	110	108	0
+ Partisanship	158	.6	.8	.6	1.5	90	42.0	12	10,000	100	112	48
+ Metropolitan	146	.5	.7	.6	1.3	93	67.8	12	10,000	138	152	102

where suffixes $i, j = 1$ denote the Conservative candidate; $i, j = 2$ denote Labour; $i, j = 3$ denote the Alliance, and $i = 4$ denote the rest in 1983. The voter transition matrix for this base model (see Table 3) indicates higher loyalty among Labour supporters, with some switching of Alliance voters to Labour. As expected in this low turnout election, previous nonvoters stayed away from the polls.

The measures of fit indicate that the model fits reasonably well with a high degree of constancy of voting behavior, but there is room for improvement. The α parameters are low, indicating considerable variation about the fitted probabilities in each row.

Examination of the correlations of the residuals for each option separately with a battery of socioeconomic and political variables indicates that the constant model has a consistent tendency to underestimate the numbers of people in the Rest category (primarily nonvoters) in constituencies characterized by relatively unstable electorates. These are constituencies in which there has been a relatively high degree of migration and those with aging electorates, resulting in a disproportionately high loss from the electoral register because of death. Note that the mean error (Measure 1) for the Rest is high in relation to the party voting options. Consequently we introduce the first covariate, Turnover, which attempts to measure the degree of instability of the electorate defined as follows: Z_{ik} = proportion of electors who have moved into constituency k in the one-year period 1980–1981 + .05 $\times$ the proportion of the electorate aged 60+ in 1981.

The factor .05 is an estimate, supplied by the British Office of Censuses and Surveys, of the death rate of people aged 60+ in a one-year period.

A more refined measure of turnover would also include emigration and new voters coming onto the register in a one-year period, but these data are unavailable.

The turnover covariate is interpreted as a concomitant variate for controlling the varying degrees of instability of the electorates rather than as a determinant of voting behavior; its inclusion casts doubt on the adequacy of the modification proposed by Hawkes. Twice the difference in log-likelihood for this model compared to the constant model is 63.6 on 12 df, so the covariate does provide a significant explanation of variation in P_k .

The third model introduces the explanatory variate Partisan-

Table 3. Voter Transition for Constant Model (1983)

Party	Conservative	Labour	Alliance	Rest
Conservative	.414	.000	.023	.563
Labour	.000	.506	.000	.494
Alliance	.000	.071	.286	.642
Rest	.003	.000	.000	.997

ship, defined as $Z_{i2k} = X_{ik}/X_{.k}$, the proportion of the 1983 electorate for option i in constituency k . Thus the transition probability for a voter who chose a particular option in the 1983 election is influenced by the existing degree of support for that option in the constituency. Such effects have been noted by several analysts of British elections (notably Miller 1977), and the rationale for them is that voters are influenced by the voting choice of their relations, colleagues, and neighbors. The shares of the electorate in the 1983 election exhibit positive correlations with their respective residuals from the Constant + Turnover model. The likelihood criterion again indicates that this variate provides a significant explanation of variation in P_k .

The final model introduces the variate Metropolitan to test the hypothesis that the residence in a metropolitan county affected voting behavior. This is defined as follows: $Z_{3k} = 1$ if constituency k is completely contained in a metropolitan county, 0 otherwise. Thus the final model fitted, following (2.8), is

$$\log(P_{ijk}/P_{ijk}) = B_{ijo} + B_{ij1}Z_{1k} + B_{ij2}Z_{2k} + B_{ij3}Z_{3k}.$$

The inclusion of the Metropolitan variate again significantly improves the fit. Note that the α parameters are all large in the final model, indicating less random variation than the other models. The second, chi-squared, term of (3.2) is 198 on 146 df for this final fitted model, whereas for an aggregated multinomial model with θ_i set to zero, the chi-squared term is 75,104 on 150 df, indicating the definite need for an allowance for overdispersion.

Table 4 displays the estimated parameters of the final model transformed back to P_k for particular values of the covariates. The first row in each subtable gives the estimated P_k for all covariates at their mean values. The second row gives the estimates for partisanship evaluated at one standard deviation above the mean, with the other variates at their mean, and shows increased loyalty among the Conservatives and the Al-

Table 4. Final Model-Selected Transition Probabilities

1983 General Election	1984 European Election				
	Conserva- tive	Labour	Alliance	Rest	Election
Conservative	.333	.001	.064	.602	Mean
	.349	.000	.071	.581	Partisanship
	.378	.000	.068	.554	Metropolitan
Labour	.000	.498	.000	.502	Mean
	.000	.473	.000	.527	Partisanship
	.000	.424	.000	.576	Metropolitan
Alliance	.000	.081	.195	.724	Mean
	.000	.090	.237	.674	Partisanship
	.056	.219	.225	.500	Metropolitan
Rest	.001	.000	.000	.999	Mean
	.001	.000	.000	.999	Partisanship
	.000	.000	.000	1.000	Metropolitan

Table 5. *Final Model: Constituencies With Largest Standardized Residuals*

Constituency	Conservative	Labour	Alliance	Rest
Bristol	2.57	2.23	-.21	-2.56
London West	1.81	1.87	.94	-1.72
Cumbria and Lancashire North	.61	.01	2.63	-1.38
Merseyside West	-.01	-2.56	.74	1.58

liance. The third row gives estimates for metropolitan constituencies with other variates at their mean. The main feature to note here is the increased switching of Alliance voters to the other main parties, particularly Labour.

Table 5 lists the four constituencies with the largest residuals after fitting the final model—they have been the largest sum of squared residuals totaled over the four options. Special local factors can be suggested to explain the deviances in three of these. These are small in terms of unstandardized residuals (e.g., the largest for the whole set of constituencies and voting options is 4% overprediction of abstention in Bristol), so a deviant result in a component constituency in 1983 may well account for the differences. Students of British politics would suggest that the Bristol constituency result shows the effect of a prominent left-wing Labour candidate in the Bristol West seat in 1983 and that the Cumbria result is affected by the recent by-election in the Penrith Westminster constituency, where the Alliance made a dramatic improvement. Merseyside West presents a specially interesting case. Here the local Labour-controlled council, which has the Alliance as its main opposition, was in dispute with the Conservative government. The final model overpredicts both turnout and the Labour vote with Alliance as the main beneficiary. The residuals suggest a greater abstention among Labour supporters and a switch to the Alliance against the national trend.

There are few large correlations of the candidate explanatory variates tested with the residuals for the final model. The largest is a correlation of .45 for the percentage of people aged 60 and over, with the Conservative residual indicating that old people may be relatively more Conservative. Similarly the percentage of people employed in distribution and catering has a correlation of .37 with the Conservative residual. All other correlations are less than .30 with most less than .10. It is interesting to note that the correlations of residuals with structural variables such as marginality and third-place party are all very low, indicating that voters were not influenced by tactical considerations in their voting choice but voted on national party lines.

The larger correlations indicate that there may yet be scope

for the addition of further terms to the model, but the data are limited and our chosen model fits well.

5.4 Discussion

A persistent feature of the fitted transition matrices obtained is the zero fitted values in some cells. Of course, any fitted model is never more than a good approximation to reality, and the question is whether zero is a poor approximation. The only comparative micro-level data that we were able to find were those of a large poll conducted for Independent Television News (ITN) on European election day (P. McKee, personal communication, 1985). The poll covered a sample of voters leaving polling stations, and transitions are based on recalled vote at the previous general election and declared vote in the European election. The poll, whose voter transition matrix is given in Table 6, was accurate in its prediction of the vote percentage for each party, with column total percentages of 39.5, 36.6, 20.8, and 3.0, compared with the true 40.5, 36.5, 19.5, and 3.5 for Conservative, Labour, Alliance, and Other, respectively.

For estimation of transitions, however, the poll is not ideal for our purposes, since it (a) suffers from internal problems of recall discussed earlier, (b) deals only with voters, not electors, and does not allow direct estimation of transitions to abstention, (c) represents a selection of polling stations across Scotland and Wales in addition to England, whereas our estimates are for England only, and (d) has too broad a categorization for "Did not vote," including those who were too young to vote and refusals as well as those who abstained in 1983. The 4% who did not vote in the poll must thus be regarded as an extreme upper bound, 2% probably being a more realistic figure. The "Others" are predominantly Nationalists and report voting choices available only in Scotland and Wales and can be ignored for comparison with our English transition estimates, following point (c).

The only comparison that we can make about transitions to abstention [point (b)] is that between the composition in terms of origin (1983 voting choice) of the actual voters in the 1984 European election obtained from the two sources. For the aggregate data, using the mean estimate figures in Table 4, we expect the English voters in the European election to comprise $33.3 \times .398$ Conservatives in 1983, $19.5 \times .498$ Labour, $19.2 \times .276$ Alliance, and $28.1 \times .001$ from previous abstainers and others, where the first figure in each pair is the proportion of the 1983 electorate and the second is the estimate of the transition rate to abstention. Scaling these figures to sum to 100% gives an expected voter profile of 46.8% Conservatives,

Table 6. *Voting Frequencies for the ITN Poll*

1983 General Election	1984 European Election					
	Conservative	Labour	Alliance	Other	Refuse	Total
Conservative	1,326	69	152	18	2	1,567
Labour	28	1,095	76	19	3	1,221
Alliance	46	109	491	23	3	672
Other	2	9	11	43	—	54
Did not vote	45	59	33	6	2	145
Total	1,447	1,341	763	109	10	3,670

Table 7. Voting Transitions From ITN Poll,
Assuming Fitted Turnout by Party

1983 General Election	1984 European Election			
	Conservative	Labour	Alliance	Rest
Conservative	.341	.018	.039	.602
Labour	.012	.455	.031	.502
Alliance	.020	.047	.209	.724
Rest	.000	.001	.000	.999

34.3% Labour, 18.7% Alliance, and .017 from abstention and others. The corresponding voter profile estimated from the poll is 43.8, 34.0, 18.3, and 3.9; these are the row total percentages in Table 6, with respondents in the Refuse and Other columns omitted. The only notable discrepancies are a 3% surplus for Conservatives and a corresponding discrepancy in the "Did not vote" figure. We can thus with some justification assume the turnout rates of Table 4 and construct from Table 6 the full transition estimates for the poll in Table 7 with these assumed turnout rates. The differences between the two tables of transition rates are not large; since Table 7 cannot be regarded as estimates of the true rates due to the provisos about the coverage and quality of the poll, we leave the detailed comparison here.

Undoubtedly zero transition rates are too low but probably not much too low. The quality of the European/General election aggregate data available can be cited as one cause. Data with marginals based on the same individuals are likely to produce truer estimates. The heterogeneity of our data is reflected in the α parameters, and these are by no means all as large as the 1,000 or so we would like [see Wilks (1962) for variance properties of the Dirichlet]. Furthermore, Section 4 points to the information provided by aggregate data and the sort of trading between transition estimates that can result from insufficient variability in the 1983 vote proportion margins across the 66 English constituencies. This could be reflected in rather complicated multidimensional confidence regions, but these are outside the scope of this article. Suffice it to say that the high level of aggregation in the Euro-constituency data works to reduce such variability in the 1983 margins.

Finally, we emphasize the diagnostics for a good model. As discussed before, the α parameters and the residuals provide the main ingredients to evaluate the fit of a model. Here the α parameters and the residuals provide the main ingredients to evaluate the fit of a model. Here the α parameters indicated a persistent but moderate level of heterogeneity or random effects. However, the residuals displayed no systematic features unaccounted for when examined both by themselves and against various candidate predictors.

APPENDIX A: FIRST DERIVATIVES OF THE AGGREGATED COMPOUND MULTINOMIAL LIKELIHOOD

We wish to derive $\partial l / \partial \mathbf{P}_k$, $\partial l / \partial \theta_i$ as given by (3.3) and (3.4) from l obtained from (3.2). For simplicity of notation we suppress the k suffix and let $\langle \mathbf{A} \rangle$ denote the (t, m) th element of the matrix $\mathbf{A}$ so that, for example, $\sum \langle \mathbf{A} \rangle$ denotes the sum of all of the elements of $\mathbf{A}$. This notation follows that of Dwyer (1967). Throughout this Appendix, $\mathbf{M}$ denotes the $(J-1) \times (J-1)$ matrix with all elements zero except the (t, m) th element, this being unity.

Lemma 1. $\partial \log |\Gamma| / \partial \mathbf{P} = \sum \langle \Gamma^{-1} \rangle \partial \langle \Gamma \rangle / \partial \mathbf{P}$. [See Dwyer (1967) for a proof of this.]

Lemma 2.

$$\begin{aligned} \partial \langle \Gamma \rangle / \partial \mathbf{P} &= -\text{diag}(\mathbf{w})\mathbf{P}\mathbf{M}^T - \text{diag}(\mathbf{w})\mathbf{P}\mathbf{M}, & t \neq m \\ &= \mathbf{w}\mathbf{u}^T - 2 \text{diag}(\mathbf{w})\mathbf{P}\mathbf{M}, & t = m, \end{aligned}$$

where $\mathbf{u}$ is a $(J-1) \times 1$ vector with m th element 1, the rest zero. This follows from first principles or by using equation (7.13) of Dwyer (1967).

Lemma 3. From Lemmas 1 and 2,

$$\partial \log |\Gamma| / \partial \mathbf{P} = \mathbf{w}\mathbf{1}^T \mathbf{D}(\Gamma^{-1}) - 2 \text{diag}(\mathbf{w})\mathbf{P}\Gamma^{-1},$$

where $\mathbf{D}(\mathbf{A})$ is a matrix that has the diagonal elements of $\mathbf{A}$ with all off-diagonal elements zero, and $\mathbf{1}$ is a $(J-1) \times 1$ vector of ones.

In order to differentiate the second term in (3.2) with respect to $\mathbf{P}$, let

$$\mathbf{r} = (\mathbf{y} - \mathbf{P}^T \mathbf{x})^T \Gamma^{-1} (\mathbf{y} - \mathbf{P}^T \mathbf{x}). \quad (\text{A.1})$$

Lemma 4. $\partial \mathbf{r} / \partial \mathbf{P} = 2\mathbf{x}(\mathbf{y} - \mathbf{P}^T \mathbf{x})^T \Gamma^{-1} + \mathbf{A}$, with $\mathbf{A} = (a_{tm}) = \langle \langle \mathbf{A} \rangle \rangle$, $I \times (J-1)$, and $\langle \mathbf{A} \rangle = (\mathbf{y} - \mathbf{P}^T \mathbf{x})^T [\partial \Gamma^{-1} / \partial \langle \mathbf{P} \rangle] (\mathbf{y} - \mathbf{P}^T \mathbf{x})$. Proof follows from differentiating as a product of three terms.

Lemma 5. $\partial \Gamma^{-1} / \partial \langle \mathbf{P} \rangle = -\Gamma^{-1} [\partial \Gamma / \partial \langle \mathbf{P} \rangle] \Gamma^{-1}$. This standard result follows from differentiating the identity $\Gamma \Gamma^{-1} = I$.

Lemma 6. $\partial \Gamma / \partial \langle \mathbf{P} \rangle = (a_{jg})$, where $a_{jg} = 0$ (j or $g \neq m$), $a_{mj} = a_{jm} = -w_j p_{ij}$ ($t \neq m$), and $a_{mm} = w_j(1 - p_{tm})$. Proof is from first principles.

Lemma 7. Combining Lemmas 4, 5, and 6 one sees that

$$\begin{aligned} \partial \mathbf{r} / \partial \mathbf{P} &= -2\mathbf{x}(\mathbf{y} - \mathbf{P}^T \mathbf{x})^T \Gamma^{-1} - \mathbf{w}[\Gamma^{-1}(\mathbf{y} - \mathbf{P}^T \mathbf{x})]^T \\ &\quad + 2 \text{diag}(\mathbf{w})\mathbf{P}\Gamma^{-1}(\mathbf{y} - \mathbf{P}^T \mathbf{x})(\mathbf{y} - \mathbf{P}^T \mathbf{x})^T \Gamma^{-1}, \end{aligned}$$

where $[\mathbf{a}]^2 = (a_1^2, \dots, a_{J-1}^2)^T$.

Finally, to differentiate l with respect to $\theta_i = 1/(1 + \alpha_i)$, the following lemmas may be used.

Lemma 8.

$$\begin{aligned} \partial \langle \Gamma \rangle / \partial \theta_i &= -p_{ti} p_{tm} x_i (x_i - 1), & t \neq m \\ &= p_{tm} (1 - p_{tm}) x_i (x_i - 1), & t = m. \end{aligned}$$

Lemma 9. Using Lemma 1 with $\mathbf{P}$ replaced by θ_i and Lemma 8,

$$\partial \log |\Gamma| / \partial \theta_i = x_i (x_i - 1) \text{trace}(\Gamma^{-1} \Omega_i),$$

where following (2.2), $\Omega_i = \text{diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^T$.

Lemma 10. Using Lemmas 5 and 8 with $\mathbf{r}$ defined by (A.1),

$$\partial \mathbf{r} / \partial \theta_i = -(\mathbf{y} - \mathbf{P}^T \mathbf{x})^T \Gamma^{-1} \Omega_i \Gamma^{-1} (\mathbf{y} - \mathbf{P}^T \mathbf{x}) x_i (x_i - 1).$$

Lemmas 9 and 10 give Equation (3.4).

APPENDIX B: AGGREGATED COMPOUND MULTINOMIAL LIKELIHOOD EXPECTED SECOND DERIVATIVES

By considering $E(-\partial^2 l / \partial \langle \mathbf{P} \rangle \partial \mathbf{P})$ we are able to simplify the expressions for the second derivatives of the log-likelihood, since $E(\mathbf{Y} - \mathbf{P}^T \mathbf{x}) = \mathbf{0}$ and $E(\mathbf{Y} - \mathbf{P}^T \mathbf{x})(\mathbf{Y} - \mathbf{P}^T \mathbf{x})^T = \Gamma$.

Differentiating (3.3) with respect to the (t, m) th element of $\mathbf{P}$, that is, $\langle \mathbf{P} \rangle$, using Lemma 5 and suppressing the k suffix, with $\mathbf{M}$ defined as in Appendix A and with $\mathbf{H} = \Gamma^{-1}$,

$$\begin{aligned} -2E(\partial^2 l / \partial \langle \mathbf{P} \rangle \partial \mathbf{P}) &= \mathbf{w}\mathbf{1}^T \mathbf{D}(-\mathbf{H}[\partial \Gamma / \partial \langle \mathbf{P} \rangle] \mathbf{H}) \\ &\quad - 2 \text{diag}(\mathbf{w})[\mathbf{M}\mathbf{H} - \mathbf{P}\mathbf{H}(\partial \Gamma / \partial \langle \mathbf{P} \rangle) \mathbf{H}] \\ &\quad + 2\mathbf{x}\mathbf{x}^T \mathbf{M}\mathbf{H} - \mathbf{w}\mathbf{1}^T E[\mathbf{D}(\mathbf{H}\{\mathbf{Y} - \mathbf{P}^T \mathbf{x}\}) \\ &\quad \times \{\mathbf{Y} - \mathbf{P}^T \mathbf{x}\}^T \mathbf{H}] / \partial \langle \mathbf{P} \rangle \\ &\quad + 2 \text{diag}(\mathbf{w})[\mathbf{M}\mathbf{H} - 2\mathbf{P}\mathbf{H}(\partial \Gamma / \partial \langle \mathbf{P} \rangle) \mathbf{H}], \end{aligned}$$

since $[a]^2 = (a_1^2, \dots, a_{j-1}^2)^T = \mathbf{1}^T \mathbf{D}(\mathbf{a}\mathbf{a}^T)$. Simplifying further,

$$-2E(\partial^2 l / \partial \langle \mathbf{P} \rangle \partial \langle \mathbf{P} \rangle) = \mathbf{w} \mathbf{1}^T \mathbf{D}(\mathbf{H} \{ \partial \Gamma / \partial \langle \mathbf{P} \rangle \} \mathbf{H}) - 2 \text{diag}(\mathbf{w}) \mathbf{P} \mathbf{H} (\partial \Gamma / \partial \langle \mathbf{P} \rangle) \mathbf{H} + 2 \mathbf{x} \mathbf{x}^T \mathbf{M} \mathbf{H}, \quad (\text{B.1})$$

and Lemma 6 can be used to express $\partial \Gamma / \partial \langle \mathbf{P} \rangle$. The first two terms are of order 1 and small compared to the third term unless θ is near 1 (α near zero). Certainly in the multinomial case, $\alpha \rightarrow \infty$, the third term is overwhelmingly dominant.

Differentiating (3.4) with respect to θ , and taking expectations leads to

$$E(-2\partial^2 l / \partial \theta_i \partial \theta_i) = -\sum x_{ik}(x_{ik} - 1) \text{trace}(\mathbf{\Omega}_{ik} \mathbf{H}_k \mathbf{\Omega}_{ik} \mathbf{H}_k),$$

using Lemmas 9 and 10.

Finally, differentiating (3.4) with respect to $\langle \mathbf{P} \rangle = p_{tmk}$ gives, suppressing k ,

$$\begin{aligned} E(\partial^2 l / \partial \theta_i \partial \langle \mathbf{P} \rangle) &= x_i(x_i - 1) [\partial \text{trace}(\mathbf{H} \mathbf{\Omega}_i) / \partial \langle \mathbf{P} \rangle] \\ &\quad + 2 \text{trace}(\mathbf{H} (\partial \Gamma / \partial \langle \mathbf{P} \rangle) \mathbf{H} \mathbf{\Omega}_i) \\ &\quad - \text{trace}(\mathbf{H} \partial \mathbf{\Omega}_i / \partial \langle \mathbf{P} \rangle) \\ &= 2x_i(x_i - 1) \text{trace}(\mathbf{H} (\partial \Gamma / \partial \langle \mathbf{P} \rangle) \mathbf{H} \mathbf{\Omega}_i). \end{aligned}$$

Lemma 6 may be used to replace $\partial \Gamma / \partial \langle \mathbf{P} \rangle$.

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